

Project 2: Datengetriebene Entscheidungsprozesse durch integrierte BI- und ML-Ansätze

Literature review:

Kriterien zur Qualitätssicherung:

- Bandara et al., 2015: Achieving Rigor in Literature Reviews

Components of a rigorous literature review methodology (to ensure quality):

1. Structured Four-Phase Approach:

1. **Phase 1**: Define scope/goals, select sources, and develop search strategies.

(Ziele und Suchstrategie festlegen sowie geeignete Quellen auswählen.)

2. **Phase 2**: Organize literature, **design coding schemes**, prepare for analysis.

Coding schemes are like categories or labels you create to help you make sense of the information you read in your papers.

(Literatur organisieren, Kodierungsschema entwerfen und Analyse vorbereiten.)

3. **Phase 3**: Analyse and synthesize literature using qualitative methods (e.g. **thematic coding**).

Thematic coding, also called thematic analysis, is a type of qualitative data analysis that finds themes in text by analyzing the meaning of words and sentence structure.

(Literatur mit qualitativen Methoden (z. B. Thematic Coding) analysieren und zusammenfassen.)

4. **Phase 4**: Report findings clearly and visually, linked to evidence.

(Ergebnisse klar und visuell darstellen und mit Belegen verknüpfen.)

2. Use of Tools for Rigor and Traceability:

Tools like **NVivo**, **ATLAS.ti**, **MAXQDA** for coding and analysing literature.

* Excel can also support structured data extraction and synthesis.

3. Inductive vs. Deductive Coding:

Deductive = We already know what we're looking for.

Inductive = We're open to discovering new, unexpected things while reading.

(Induktive vs. deduktive Kodierung)

For our project, a **mixed approach** (deductive categories like “BI”, “ETL”, “ML” + inductive themes from the texts) is ideal.

(Ein kombinierter Ansatz aus vorab festgelegten Themen und offenen Kategorien ist sinnvoll).

4. Tool-Aided Transparency:

We should keep **memos, annotations, and coding logs** during analysis for traceability.

A **coding log** is like a **diary or notebook** where we write down:

- What codes (categories) we used
- Why we used them
- What decisions we made while reading and coding papers

(Notizen, Kommentare und Kodierungsprotokolle erhöhen die Nachvollziehbarkeit.)

We should ensure **inter-coder reliability** if working in a team.

(Bei Teamarbeit sollte die Übereinstimmung zwischen Kodierenden überprüft werden.)

If two (or more) people are coding the same texts, **inter-coder reliability** shows how much agreement there is between different people.

5. Transparent Documentation:

(Transparente Dokumentation des gesamten Prozesses)

Document each step clearly: how sources were found, how inclusion/exclusion criteria were applied, and how findings were synthesized.

(Jeder Schritt sollte genau dokumentiert werden.)

- Cram et al., 2020 – (Re)considering Literature Review Reproducibility:

1.Systematicity:

Follow structured, step-by-step procedures (search, screening, coding, analysis).

Adopt or adapt frameworks like **Okoli's** or Fink's steps.

Okoli's steps:



2.Transparency:

Clearly report **how** our review was conducted: sources used, search terms, criteria, decisions.

Helps others evaluate and trust our results.

3.Reproducibility vs. Replicability:

Reproducibility: Others can repeat our *methods* (e.g. same search and screening process).

Replicability: A new study could independently validate our results.

For our project, **reproducibility of the review process** is key for quality assurance.

4.Desirability of Reproducibility:

Particularly important in **theory-testing** or **tool-assessing** reviews like ours.

Ensures future teams or reviewers can **audit or update** our work systematically.

Summary

Criterion	Requirement
Systematic Search	Clear strategy, keywords, databases, inclusion/exclusion (
Transparent Documentation	Step-by-step protocol (methods, tools, selection logic)
Tool-Supported Coding	Use NVivo/Excel/Atlas.ti for traceable synthesis
Reproducibility	Others can repeat your method and get similar insights
Iterative Refinement	Allow codebook and themes to evolve with findings
Clear Reporting	Visual summaries, logic diagrams, dashboards